

IBM Data Analyst Capstone Project Report

- ▶ Technology Trends Analysis Dashboard Using IBM Cognos Analytics
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- ▶ Date: 27-05-2026

Executive Summary

- Analyzed the Stack Overflow Developer Survey dataset containing over 18,000 responses.
- Used IBM Cognos Analytics, Python, SQL, and data visualization techniques to identify current technology trends and future technology preferences.
- Developed interactive dashboards to analyze programming languages, databases, platforms, web frameworks, and demographics.
- Key findings showed strong adoption of modern programming languages and increasing interest in cloud and emerging technologies.
- Insights from the analysis can support workforce planning, skill development, and technology investment decisions.

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Introduction

The purpose of this project was to analyze developer survey data to understand:

- ▶ Current technology usage trends
- ▶ Future technology preferences
- ▶ Demographic distribution of developers
- ▶ Technology adoption patterns

The project demonstrates how data analytics and visualization can transform raw survey data into actionable business insights.

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Business Problem

Organizations need data-driven insights to:

- Identify high-demand technologies
- Understand future technology trends
- Plan workforce development strategies
- Align employee training with market demand
- Support strategic technology investments

This project aims to provide analytical insights using developer survey data.

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Dataset Overview

Dataset: Stack Overflow Developer Survey Records: Approximately 18,845 responses
File Used: survey_data_updated.csv

Major fields analyzed:

- Programming Languages
- Databases
- Platforms
- Web Frameworks
- Country
- Age
- Education Level

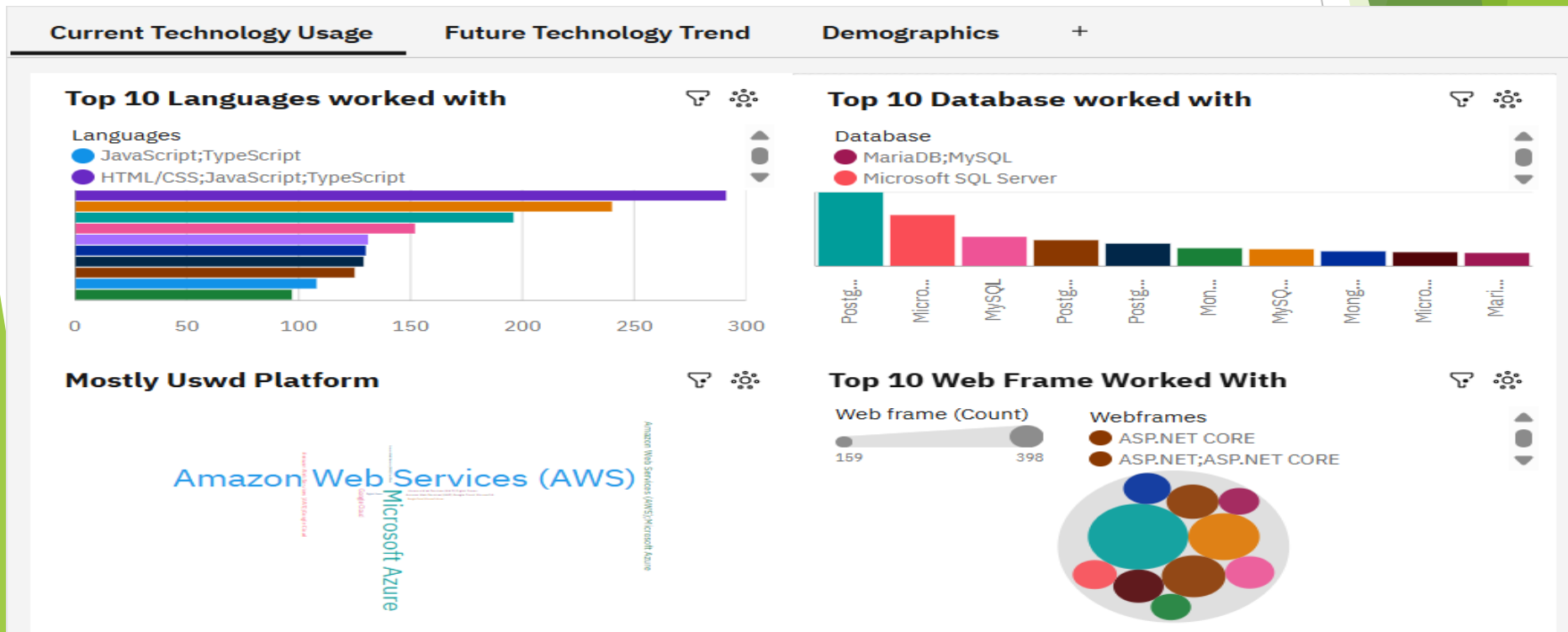
Methodology

- ▶ • Dataset: survey_data_updated.csv (18,845 responses)
- ▶ • Data preparation and filtering
- ▶ • Dashboard development in IBM Cognos Analytics
- ▶ • Visual analysis using bar charts, column charts, word clouds, tree maps, maps, line charts, and stacked bar charts

Results - Current Technology Usage

Dashboard visualizations:

- Top 10 Languages Used
- Top 10 Databases Used
- Top 10 Platforms Used
- Top 10 Web Frameworks Used



Current Technology Usage Findings

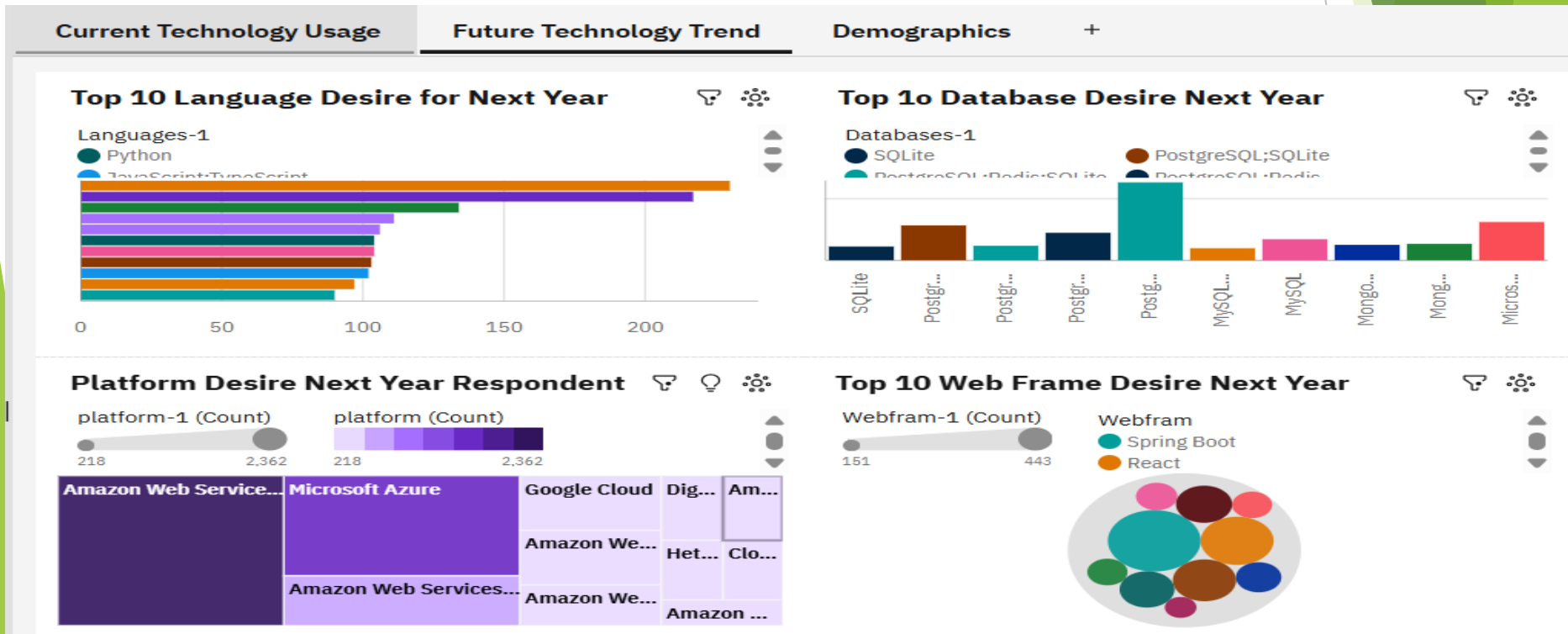
Key insights from the dashboard:

- JavaScript and Python emerged among the most widely used programming languages.
- PostgreSQL and MySQL showed strong database adoption among developers.
- Cloud and web-based platforms demonstrated significant usage trends.
- Modern web frameworks continue to dominate development ecosystems.
- Technology adoption patterns indicate strong preference toward scalable and open-source tools.

Results - Future Technology Trends

Dashboard visualizations:

- Top 10 Desired Languages
- Top 10 Desired Databases
- Top 10 Desired Platforms
- Top 10 Desired Web Frameworks



Future Technology Trends Findings

Key insights from the dashboard:

- Developers showed increasing interest in modern and emerging programming languages.
- Cloud technologies and modern platforms remain highly desired.
- PostgreSQL and cloud-native databases demonstrated strong future demand.
- Interest in modern web frameworks indicates continued growth in full-stack development.
- Technology trends suggest a shift toward scalable, cloud-based, and data-driven solutions.

Results - Demographics

Dashboard visualizations:

- Age Distribution
- Respondent Count by Country
- Education Level Distribution
- Age vs Education Level

Current Technology Usage

Future Technology Trend

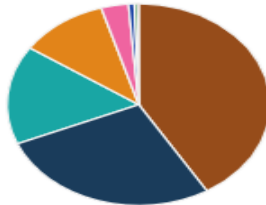
Demographics

+

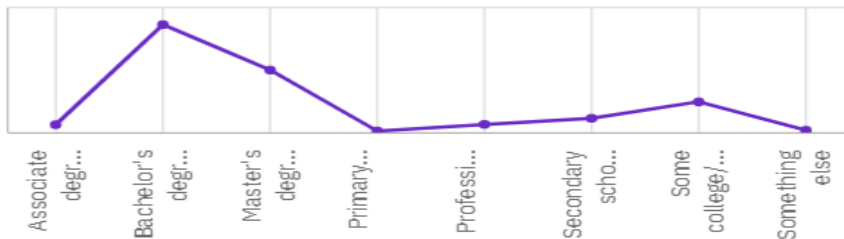
Respondent distribution by Age

Age

- 25-34 years old
- 35-44 years old
- 18-24 years old
- 45-54 years old



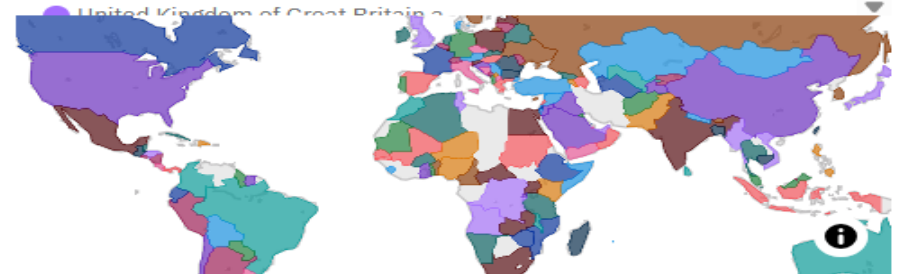
Respondent Distribution by Formal Education Level



Respondent Count by Country

Country

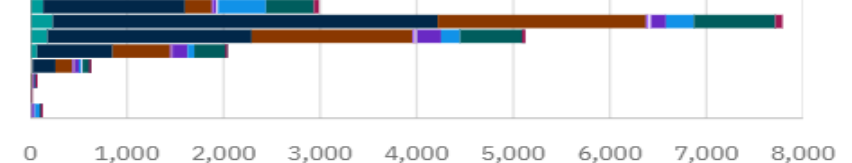
United Kingdom of Great Britain



Respondent Count by Age and Education Level

Education Levels

- Associate degree (A.A., A.S., etc.)
- Bachelor's degree (B.A., B.S., B.F.)



Demographics Findings

Key insights from the dashboard:

- ▶ Most respondents belonged to the early and mid-career age groups.
- ▶ The survey represented developers from diverse geographic regions.
- ▶ Bachelor's and Master's degree holders formed a major portion of respondents.
- ▶ Education level influenced technology preferences and career trends.
- ▶ Demographic diversity highlights the global nature of the developer community.

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Overall Technology Trends

Summary of observed trends:

- ▶ Strong adoption of modern programming languages and cloud technologies.
- ▶ Increasing interest in scalable and open-source platforms.
- ▶ Growing demand for data-driven technologies and analytics tools.
- ▶ Continuous evolution of web development frameworks and databases.
- ▶ Technology preferences vary across demographics and regions.

Business Insights

Business value generated from the analysis:

- ▶ Helps organizations identify high-demand technical skills.
- ▶ Supports strategic hiring and workforce planning.
- ▶ Assists in technology investment and adoption decisions.
- ▶ Enables data-driven learning and development initiatives.
- ▶ Provides insight into future industry technology directions

Recommendations

Recommendations based on the analysis:

- ▶ Invest in cloud computing and data analytics skill development.
- ▶ Encourage learning of modern programming languages and frameworks.
- ▶ Focus on scalable database and platform technologies.
- ▶ Monitor emerging technology trends for competitive advantage.
- ▶ Use demographic insights to support targeted training strategies.

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Discussion

The project demonstrated how interactive dashboards and visual analytics can uncover patterns in large-scale survey datasets.

- ▶ The analysis highlighted:
- ▶ Current technology adoption
- ▶ Future technology interests
- ▶ Demographic influences on technology trends

These findings can support informed business and workforce decisions.

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Conclusion

- ▶ The IBM Data Analyst Capstone Project successfully transformed raw survey data into meaningful insights using data analysis and visualization techniques.
- ▶ IBM Cognos Analytics dashboards enabled effective communication of technology trends, demographic patterns, and future technology interests.
- ▶ The project demonstrates the importance of data-driven decision making in modern organizations.

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Appendix

Additional information:

- ▶ Dataset: survey_data_updated.csv
- ▶ Tool Used: IBM Cognos Analytics
- ▶ Total Responses: ~18,845
- ▶ Guidelines by IBM

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Thank You

By - Anurag Kumar Poddar